

KS3: Water and Sanitation Through Time

"Why did it take so long to clean up Britain?"

MASTERY PACK: COMPLETE UNIT

Operative / Student Name: _____

Target: Total Recall of 126 Crucial Questions



⚡ Interactive Recall Quiz & Scratch-Off Vault

Scan with your phone to launch Instant-Feedback Recall Quiz (#practice-mode) & tap-to-reveal self-marking solutions.

meoncross-history.netlify.app/units/water_and_sanitation/mastery_pack_full.html#practice-mode

RESTRICTED ACCESS - MASTER YOUR MEMORY

How to Hack Your Memory

(Rules of Engagement)

1. Never Just Read

Reading the answers feels like learning, but it's an illusion. You must force your brain to struggle to find the answer. The struggle is what builds memory pathways.

2. Cover, Recall, Check

Cover the answer sheet. Write your answer down in the notes section (or say it out loud). Only then are you allowed to check 'The Vault'.

3. Track Your Threat Level

Use the R-A-G (Red, Amber, Green) checkboxes next to every question after you attempt it.

- ☐ **Green:** I nailed it instantly.
- ☐ **Amber:** I got it, but I had to think hard.
- ☐ **Red:** My mind went blank. (Revisit these tomorrow).

4. The Interrogation (Homework Protocol)

True mastery requires testing under pressure. Hand 'The Vault' (answers) to your parent or guardian. They will test you on 20 random questions. They hold the score; they hold the signature.

5. The 3-Star Parent Effort Rubric

When authenticating your spaced retrieval practice on the final Mastery Tracker, your parent or guardian will evaluate your effort using the following criteria:

- **★ ★ ★ 3 Stars (Full Mastery):** Completed under strict exam conditions without glancing at notes or the Knowledge Vault. Total independent recall.
- **★ ★ 2 Stars (Consolidating):** Required 1–2 hints from the Knowledge Vault, but demonstrated sound foundational understanding.
- **★ 1 Star (Needs Intervention):** Rushed, mind went blank, or completed with notes open. Target facts must be hunted down and re-tested tomorrow.

How much progress did the Romans make in public health?

Q#	The Target Question	R	A	G	Notes / My Answer
1	Why was digging cesspits in the Iron Age a hygienic way to manage waste?	☐	☐	☐	
2	What technology did Roman engineers use to bring fresh water over miles into British towns?	☐	☐	☐	
3	Which Roman fort on Hadrian's Wall contains Britain's best-preserved communal latrine, with running water channels flushing waste into sewers?	☐	☐	☐	
4	What did Roman soldiers use to wipe themselves in communal latrines?	☐	☐	☐	
5	Which Emperor ordered the construction of a famous 73-mile defensive wall with fort bathhouses across northern Britain?	☐	☐	☐	
6	What was the primary purpose of Roman public aqueducts?	☐	☐	☐	
7	In a Roman town, where did wastewater from public baths and latrines drain?	☐	☐	☐	
8	What did wealthy Romans use to heat their private villas and bathhouses?	☐	☐	☐	
9	Why did the Romans build public bathhouses in almost every military fort and town?	☐	☐	☐	
10	What instrument did Romans use to clean their skin in the bathhouse instead of soap?	☐	☐	☐	
11	What happened to Roman water and sanitation infrastructure after the legions left Britain in 410 AD?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
12	Why did Iron Age Britons have fewer sanitation crises before the Roman invasion?	☐	☐	☐	
13	What material was commonly used by Roman engineers to line water conduits and cisterns to make them watertight?	☐	☐	☐	
14	What was the caldarium inside a Roman public bathhouse?	☐	☐	☐	
15	Who paid for the construction of most public bathhouses and fountains in Roman Britain?	☐	☐	☐	
16	What health hazard was unwittingly introduced by Roman plumbing in Britain?	☐	☐	☐	
17	What does AD stand for in historical dates?	☐	☐	☐	
18	What century is the year 1066 in?	☐	☐	☐	
19	Which ancient empire conquered Britain and built Hadrian's Wall?	☐	☐	☐	
20	What does the word 'Sanitation' mean?	☐	☐	☐	

MISSION DEBRIEF

Score: _____ / 20

Parent/Guardian Authentication (Signature): _____

Date: _____

How much progress did the Romans make in public health?

Q#	The Target Question	R	A	G	Notes / My Answer
21	What is a primary source?	☐	☐	☐	
22	What did Roman bathers experience inside a bathhouse like Bearsden?	☐	☐	☐	
23	Why did medieval villages like Wharram Percy avoid major public health crises despite using simple cesspits?	☐	☐	☐	
24	On the twelfth-century plans of Canterbury Priory, what did the red lines represent?	☐	☐	☐	
25	How did poorer townspeople get their fresh water if they did not own a private well?	☐	☐	☐	
26	Why were medieval Christian monasteries cleaner and healthier than towns?	☐	☐	☐	
27	What was the primary medieval theory used to explain why diseases like the Black Death spread?	☐	☐	☐	
28	What medieval occupation involved emptying city cesspits and privy vaults?	☐	☐	☐	
29	In what year did the catastrophic Black Death first arrive in Britain through the Dorset port of Melcombe Regis?	☐	☐	☐	
30	How did medieval town councils like London attempt to combat street filth?	☐	☐	☐	
31	Where did most ordinary medieval townspeople obtain their drinking water?	☐	☐	☐	
32	What did medieval people believe caused the Black Death besides miasma?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
33	Why did medieval butchers create a major public health nuisance in towns?	☐	☐	☐	
34	What was a medieval "privy"?	☐	☐	☐	
35	What proportion of the British population is estimated to have died during the 1348–1349 Black Death?	☐	☐	☐	
36	What did the 1388 Parliament of Cambridge attempt to do regarding public health?	☐	☐	☐	
37	Why did the Great Conduit in London (built in 1245) become famous?	☐	☐	☐	
38	How did the Romans transport massive amounts of fresh water?	☐	☐	☐	
39	What did the Romans use to flush their public toilets?	☐	☐	☐	
40	Why did the Roman army need to stay healthy?	☐	☐	☐	

MISSION DEBRIEF

Score: _____ / 20

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Date: _____

Why did public health decline during the Middle Ages?

Q#	The Target Question	R	A	G	Notes / My Answer
41	What did the Romans build in their forts to treat sick soldiers?	☐	☐	☐	
42	Did the Romans understand bacteria?	☐	☐	☐	
43	What material did archaeologists discover in medieval cesspits that shows how villagers wiped themselves?	☐	☐	☐	
44	Why did medieval 'gongfarmers' perform their job exclusively under the cover of darkness?	☐	☐	☐	
45	Why did London's sanitation problems get worse during the Early Modern period?	☐	☐	☐	
46	Which famous seventeenth-century Londoner wrote about his cellar being flooded by his neighbor's toilet waste?	☐	☐	☐	
47	What devastating disease struck London in 1665, killing an estimated 100,000 people?	☐	☐	☐	
48	What preventative measure did London authorities take during the 1665 Great Plague when someone fell ill?	☐	☐	☐	
49	What was Sir John Harington's 1596 invention called in his satirical book?	☐	☐	☐	
50	Why did London's rapid population growth between 1500 and 1700 make public health worse?	☐	☐	☐	
51	What did early modern Londoners believe smoking tobacco would do during the 1665 plague?	☐	☐	☐	
52	Who were "searchers of the dead" during early modern	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	plague outbreaks?				
53	What happened to London's hygiene and disease levels after the Great Fire of 1666?	☐	☐	☐	
54	What were early modern "Bills of Mortality"?	☐	☐	☐	
55	What was the Great Conduit water system supplemented by in London in 1613?	☐	☐	☐	
56	Why were early modern London cesspits often built directly underneath house floorboards?	☐	☐	☐	
57	What did early modern physicians wear to protect themselves when examining plague victims?	☐	☐	☐	
58	How did wealthy early modern citizens protect their drinking water compared to the poor?	☐	☐	☐	
59	Why were medieval towns dangerous for public health?	☐	☐	☐	
60	What was a Gongfermer?	☐	☐	☐	

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Score: _____ / 20

Parent/Guardian Authentication (Signature): _____

Date: _____

To what extent did towns become filthier during the Early Modern period?

Q#	The Target Question	R	A	G	Notes / My Answer
61	Why did medieval monks build clean water systems?	☐	☐	☐	
62	What is a cesspit?	☐	☐	☐	
63	What did King Edward III order in 1357?	☐	☐	☐	
64	Who invented the first water closet (flushing toilet) in Britain in 1596?	☐	☐	☐	
65	Why was the 1596 flushing toilet not used by ordinary citizens?	☐	☐	☐	
66	What was a 'water seller's' job in an Early Modern town?	☐	☐	☐	
67	By how much did Britain's population grow during the century of industrialisation (1750–1850)?	☐	☐	☐	
68	In what year did Asiatic Cholera first arrive on British shores in the port of Sunderland?	☐	☐	☐	
69	What nickname did the terrified Victorian public give to cholera because of its horrifying physical symptoms?	☐	☐	☐	
70	What was the central argument of Edwin Chadwick's landmark 1842 Sanitary Report?	☐	☐	☐	
71	Why were Victorian "back-to-back" houses so hazardous to public health?	☐	☐	☐	
72	What does the economic and political philosophy of "laissez-faire" mean?	☐	☐	☐	
73	Why did ratepayers (local taxpayers and landlords) vigorously oppose Chadwick's	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	sanitation proposals in the 1840s?				
74	What was the fatal flaw of the Public Health Act of 1848?	☐	☐	☐	
75	How quickly could cholera kill a previously healthy Victorian adult?	☐	☐	☐	
76	What was the average life expectancy for a working-class labourer in industrial Manchester in the 1840s?	☐	☐	☐	
77	Where did human waste go when private flush water closets were installed in wealthy Victorian homes before 1855?	☐	☐	☐	
78	Why was the 1848 General Board of Health disbanded in 1854?	☐	☐	☐	
79	What was a "court" or "yard" in an early 19th-century industrial town?	☐	☐	☐	
80	Why did rivers in early modern towns become open sewers?	☐	☐	☐	

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Date: _____

How did the Industrial Revolution lead to a public health crisis?

Q#	The Target Question	R	A	G	Notes / My Answer
81	Who invented a working flushing toilet in 1596?	☐	☐	☐	
82	Why did Harington's flushing toilet fail to catch on for 250 years?	☐	☐	☐	
83	What does 'urbanisation' mean?	☐	☐	☐	
84	How did early modern townspeople get their drinking water?	☐	☐	☐	
85	How did working-class families living in terraced yards usually obtain their water?	☐	☐	☐	
86	What was the terrifying waterborne disease that first struck Britain in 1831, killing 31,000 people?	☐	☐	☐	
87	What scientific theory did Victorian doctors believe in before they understood that cholera was spread by dirty water?	☐	☐	☐	
88	Why did cesspits under shared yard privies frequently overflow in industrial cities?	☐	☐	☐	
89	What famous environmental crisis in London during the blazing summer of 1858 forced politicians to act?	☐	☐	☐	
90	How did politicians in the Houses of Parliament try to cope with the foul fumes of the Great Stink in 1858?	☐	☐	☐	
91	Who was appointed chief engineer of the Metropolitan Board of Works to design London's new sewer system?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
92	How did Bazalgette's intercepting sewer system solve London's sewage crisis?	☐	☐	☐	
93	What remarkable design decision did Bazalgette make when sizing his underground sewer tunnels?	☐	☐	☐	
94	What type of mortar did Bazalgette use to ensure the brick sewers would not decay from water and waste acids?	☐	☐	☐	
95	What major medical breakthrough in 1861 by Louis Pasteur finally discredited the Miasma Theory?	☐	☐	☐	
96	Why is the Public Health Act of 1875 considered a massive turning point in British history?	☐	☐	☐	
97	What grand public embankments were built in London on top of Bazalgette's low-level intercepting sewers?	☐	☐	☐	
98	When cholera struck London for the final time in 1866, what did the health statistics reveal about Bazalgette's sewers?	☐	☐	☐	
99	What Victorian engineering landmarks were constructed by Bazalgette to pump sewage up to river discharge levels?	☐	☐	☐	
100	What political change in 1867 helped encourage Parliament to pass stronger public health laws like the 1875 Act?	☐	☐	☐	

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Why did it take the 'Great Stink' to finally clean up Britain's streets?

Q#	The Target Question	R	A	G	Notes / My Answer
101	What was the 'laissez-faire' attitude?	☐	☐	☐	
102	What terrifying waterborne disease first struck Britain in 1831?	☐	☐	☐	
103	What scientific theory did doctors wrongly believe caused disease?	☐	☐	☐	
104	How did Dr. John Snow prove cholera was waterborne?	☐	☐	☐	
105	Who published a damning report on sanitary conditions in 1842?	☐	☐	☐	
106	What was Dr John Snow's revolutionary hypothesis regarding how cholera was transmitted in 1849?	☐	☐	☐	
107	During the September 1854 cholera outbreak in Soho, what specific piece of evidence did John Snow create to prove his theory?	☐	☐	☐	
108	What did John Snow convince the parish Board of Guardians to do on 8 September 1854 to halt the Soho epidemic?	☐	☐	☐	
109	What was discovered when inspectors later excavated the Broad Street pump well?	☐	☐	☐	
110	Why were workers at the nearby Broad Street brewery completely unaffected by the 1854 cholera epidemic?	☐	☐	☐	
111	Why did the British medical establishment and General Board of Health initially reject John Snow's discovery?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
112	What German bacteriologist finally isolated and proved the cholera bacterium (<i>Vibrio cholerae</i>) under a microscope in 1883?	☐	☐	☐	
113	How did John Snow use the Lambeth Water Company versus Southwark & Vauxhall Water Company data to prove his waterborne theory?	☐	☐	☐	
114	Which historical period made the LEAST progress in public health engineering in Britain?	☐	☐	☐	
115	What was the primary obstacle preventing earlier public health reform throughout British history?	☐	☐	☐	
116	Which of the following was a key feature of the compulsory Public Health Act of 1875?	☐	☐	☐	
117	How did Roman public health differ fundamentally from Victorian public health?	☐	☐	☐	
118	What role did Edwin Chadwick play in the story of British public health?	☐	☐	☐	
119	What was the significance of Joseph Bazalgette's intercepting sewer system for the city of London?	☐	☐	☐	
120	Why did John Snow's investigation become famous as a founding moment of modern medical science?	☐	☐	☐	

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End of Unit Assessment: Cholera Investigation

Q#	The Target Question	R	A	G	Notes / My Answer
121	Looking back across all six lessons, why was progress in British public health NOT a simple straight line of improvement?	☐	☐	☐	
122	What Roman engineering structure carried fresh water across valleys into towns?	☐	☐	☐	
123	What was a medieval worker who cleared cesspits at night called?	☐	☐	☐	
124	What incorrect theory claimed that foul smells and poisonous air caused disease?	☐	☐	☐	
125	In what year did Asiatic cholera first strike Britain?	☐	☐	☐	
126	What 1858 environmental crisis forced Parliament to fund London's new sewer system?	☐	☐	☐	

MISSION DEBRIEF

Score: _____ / 20

Parent/Guardian Authentication (Signature): _____

Date: _____

THE VAULT

Restricted Access: Answer Keys

How much progress did the Romans make in public health?

1. Roundhouse settlements were spread out and temporary pits kept waste away from springs
2. Stone channels called conduits that utilized the natural pull of gravity
3. Housesteads Fort on Hadrian's Wall
4. A wet sponge attached to the end of a shared wooden stick
5. Emperor Hadrian
6. To transport clean spring water continuously into towns using gravity
7. Into underground stone sewers like the Cloaca Maxima that discharged into local rivers
8. A hypocaust system circulating hot furnace air beneath raised floors
9. Because bathing, exercise, and socialising were central to Roman civic culture and soldier fitness
10. A metal scraper called a strigil used with olive oil
11. Town populations collapsed and stone aqueducts fell into decay and disrepair
12. Their dispersed, small population meant waste did not build up enough to contaminate water supplies
13. Waterproof hydraulic mortar (opus signinum)
14. The hot, steamy bathing room
15. Wealthy local magistrates and benefactors seeking political status
16. Lead poisoning from lead water pipes and cooking vessels
17. Anno Domini (the year of our Lord).
18. The 11th Century.
19. The Romans.
20. Conditions relating to public health, especially clean drinking water and sewage disposal.
21. Evidence that was created at the time of the event.
22. Washing in sequence through cold, warm, and hot rooms
23. The low population density meant waste did not build up enough to contaminate water supplies
24. Pipes carrying dirty waste water away to flush the toilets
25. They bought it from water sellers who hauled river water in barrels
26. Monasteries were carefully sited near fresh rivers with dedicated leats, fresh-water lavatoria, and isolated infirmaries
27. Miasma Theory (bad air and poisonous stench)
28. Gongfermer (or Nightman)
29. 1348

- 30.** By passing municipal laws fining residents who threw waste into the street and appointing rakers
- 31.** From communal public conduits, town wells, or local water carriers called cobs
- 32.** God's wrath punishing humanity for sin, and planetary alignments
- 33.** They dumped animal blood, entrails, and offal into public streets and rivers like the Fleet
- 34.** A basic outhouse toilet built over a cesspit or watercourse
- 35.** Between 30% and 50%
- 36.** It passed an Act forbidding the dumping of dung, offal, and animal entrails into ditches and rivers
- 37.** It carried clean spring water from Tyburn into Cheapside through lead pipes
- 38.** Aqueducts.
- 39.** Running water from the aqueducts and bathhouses.
- 40.** To march and fight efficiently across the Empire.

Why did public health decline during the Middle Ages?

41. Hospitals (Valetudinaria).
42. No, they just knew bad smells or swamps were linked to illness.
43. Wild moss
44. To avoid disrupting the busy town streets with terrible smells and waste carts
45. The population grew rapidly, putting too much pressure on crowded town systems
46. Samuel Pepys
47. The Great Plague
48. They quarantined the entire household inside, painting a red cross on the door with "Lord have mercy upon us"
49. The Metamorphosis of Ajax (a pun on "a jakes", meaning a privy)
50. The city expanded into swampy marshes with no building regulations, overcrowding timber houses and overflowing cesspits
51. Ward off poisonous miasmas and prevent plague infection
52. Elderly parish women appointed to inspect corpses and report the official cause of death
53. The fire destroyed thousands of rat-infested wooden tenements, and rebuilt streets were wider and brick-built
54. Weekly published parish statistics listing total deaths and their recorded causes
55. Sir Hugh Myddelton's New River project, bringing fresh Hertfordshire spring water along a 40-mile canal
56. Because urban land was extremely scarce and houses had no gardens or backyards
57. Beaked leather masks filled with fragrant dried herbs, long waxed coats, and wide-brimmed hats
58. They bought piped water delivered through private elm-wood pipes directly to their basements a few days a week
59. They were overcrowded, with cesspits leaking into drinking wells.
60. A medieval worker who cleared human waste from cesspits at night.

To what extent did towns become filthier during the Early Modern period?

61. They believed cleanliness brought them closer to God.
62. A pit for the disposal of liquid waste and sewage.
63. The removal of filth from London's streets to prevent sickness.
64. Sir John Harington
65. Houses lacked running piped water and connections to street sewers
66. Transporting river water in large barrels on horseback to sell to homes
67. It skyrocketed from 6 million to 21 million.
68. 1831
69. King Cholera (or the Blue Death)

- 70.** That filthy living conditions caused disease, which created poverty, burdening taxpayers with higher poor rates
- 71.** They shared common side and rear walls with no through-ventilation, and multiple families shared a single privy in an unpaved court
- 72.** A policy of non-intervention where government leaves trade and public life to private individuals
- 73.** They did not want their local taxes increased to pay for expensive sewer pipes and drainage
- 74.** It was completely voluntary: councils were only forced to act if the death rate exceeded 23 per 1,000
- 75.** Within 12 to 24 hours of violent diarrhoea, vomiting, and extreme dehydration
- 76.** Under 20 years (approx. 17–19 years)
- 77.** Directly into existing cesspools underneath houses or into street gutters flowing into the Thames
- 78.** Because Edwin Chadwick's arrogant, domineering style alienated politicians and angered ratepayers
- 79.** A narrow, enclosed dead-end alley packed with back-to-back dwellings and shared privies
- 80.** Population growth outpaced the ability to dispose of waste.

How did the Industrial Revolution lead to a public health crisis?

81. John Harington.
82. There were no underground sewers to connect it to.
83. The increase in the proportion of people living in towns and cities.
84. Buying it from water sellers or sharing local pumps.
85. From a shared pump in the street or yard that only ran for a few hours.
86. Cholera
87. Miasma Theory (the belief that disease is spread by bad smells)
88. Landlords refused to pay for them to be emptied, and they were not connected to sewers.
89. The Great Stink of the River Thames
90. They soaked parliamentary window curtains in chloride of lime to neutralize the stench
91. Sir Joseph Bazalgette
92. By building 82 miles of underground brick tunnels parallel to the Thames to carry sewage east of London
93. He calculated the maximum flow needed, and then doubled the diameter to handle future population growth
94. Portland cement, tested rigorously for strength and waterproof durability
95. The publication of Germ Theory, proving microbes cause disease and fermentation
96. It ended laissez-faire by making it compulsory for local councils to provide clean water, pave streets, and inspect drains
97. The Victoria, Albert, and Chelsea Embankments along the Thames
98. Areas connected to Bazalgette's new sewers suffered virtually no cholera deaths, proving clean drainage saved lives
99. The Abbey Mills and Crossness Pumping Stations, nicknamed "Cathedrals of Sewage"
100. The Second Reform Act, which gave the vote to urban working-class men

Why did it take the 'Great Stink' to finally clean up Britain's streets?

101. The government's belief that it shouldn't interfere in people's daily lives or public health.
102. Cholera.
103. Miasma Theory (bad smells).
104. By mapping the deaths around the Broad Street water pump.
105. Edwin Chadwick.
106. It was a waterborne disease transmitted by swallowing invisible contaminated water or food
107. A street map plotting each cholera death as a black bar outside houses near the Broad Street pump
108. Remove the handle from the Broad Street water pump so residents could not drink the water

- 109. A cracked brick cesspool from house number 40 was leaking baby cholera waste directly into the well water
- 110. They were allowed free beer and had their own private deep well on site, so they never drank from the Broad Street pump
- 111. Because powerful authorities like Edwin Chadwick and Florence Nightingale were deeply committed to Miasma Theory
- 112. Robert Koch
- 113. He compared customers on the same streets: those drinking from Lambeth (intake moved upriver) had vastly lower deaths than those drinking Southwark sewage
- 114. The Early Middle Ages (500–1000 AD)
- 115. Laissez-faire government attitudes, high costs for ratepayers, and stubborn adherence to Miasma Theory
- 116. Every local council was legally required to appoint a Medical Officer of Health and a Sanitary Inspector
- 117. Romans focused on military fitness and civic bathing without knowing about bacteria; Victorians eventually developed scientific bacteriology and compulsory laws
- 118. He demonstrated statistically that poor sanitation caused illness and poverty, campaigning for the 1848 Public Health Act
- 119. It eradicated waterborne cholera from London and created a clean urban drainage infrastructure that still operates today
- 120. It pioneered the field of epidemiology — using data collection, map analysis, and scientific deduction to track and control epidemics

End of Unit Assessment: Cholera Investigation

- 121. Because Roman sanitary engineering collapsed in the Middle Ages, urbanisation overwhelmed early modern towns, and scientific understanding was delayed by false theories like miasma
- 122. An aqueduct, using the pull of gravity.
- 123. A gongfermer (or nightman).
- 124. The Miasma Theory.
- 125. 1831 (in Sunderland).
- 126. The Great Stink of the River Thames.

Mastery Tracker

Track your total recall score across all 126 questions.

Attempt	Date	Score (/126)	Parent Validation Stamp	Target for Next Time
1			PARENT VALIDATION STAMP Effort: ★ ★ ★ (Circle 1–3) Date: _____ Signature: _____	
2			PARENT VALIDATION STAMP Effort: ★ ★ ★ (Circle 1–3) Date: _____ Signature: _____	
3			PARENT VALIDATION STAMP Effort: ★ ★ ★ (Circle 1–3) Date: _____ Signature: _____	

Operative Reflection

My ultimate strength in this unit is...

The three specific facts I need to hunt down and memorize tonight are...